V9938 MSX-VIDEO

Technical Data Book

ASCII CORPORATION / NIPPON GAKKI CO., LTD.

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(Yamaha logo at bottom of page)

PREFACE

The V9938 introduced in this manual is a Very Large-Scale Integrated Circuit (VLSI) that was developed as a Video Display Processor (VDP) for the MSX2. The MSX personal computer standard was introduced in 1983 by ASCII Corporation and Microsoft Incorporated. At present, the MSX is manufactured and marketed worldwide. In 1985, out of the desire to strengthen some of the functions of the original MSX, the MSX2 standard was developed. In addition to being software-compatible with the MSX, the MSX2 supports new media and has video processing capabilities that are not available on conventional 8-bit personal computers.

To make the MSX2 a reality, two requirements for the Video Processor were upward compatibility with the existing TMS9918A (the VDP for the MSX) software while increasing the number of functions. The V9938 was developed through the joint efforts of ASCII Corporation, Microsoft Incorporated, and YAMAHA.

The following functions are supported on the V9938.

- Full bit-mapped mode
- 80-column text display
- Access using X- and Y-coordinates. The load of the I/O driver has been lightened. The X-Y coordinates are independent of the screen mode.
- Fundamental commands implemented by hardware to decrease the processing time of the I/O driver: AREA MOVE, LINE, SEARCH, RASTER OPERATION, etc.
- Digitize and external synchronization
- Color palette (9 bits x 16 patterns)
- Linear RGB video output
- More sprites per horizontal line

Because the V9938 has the above functions, it provides for superior video capabilities that make it possible for its use in a variety of applications, including the MSX2. CAPTAIN terminals and NAPLPS terminals using the V9938 have already been developed. We hope that the V9938 will be a standard video processing device on a worldwide basis.

This manual was written so as to explain how to set the parameters of the V9938 and is a reference for developing applications and systems software for it.

We are pleased that you have chosen to develop software for the V9938 and that you have referred to this manual for assistance.

Finally, we would like to express our deep gratitude to the people at NTT as well as the other related manufacturers for their valuable opinions which contributed to the development of the v9938.

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ASCII Corporation

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PART 1

MSX-VIDEO DATA PROCESSOR
V9938
USER'S MANUAL

BASIC INPUT AND OUTPUT

1. Accessing the Control Registers

There are two ways to set data in the MSX-VIDEO control registers (R#0 to R#46), which we will describe below.

1.1 Direct access

Output the data and the register number in sequence to port #1. Since this order is always used, be careful when you access the MSX-VIDEO for an interrupt routine.

	MSB 7	6	5	4	3	2	1	0 LSB	
Port #1 First byte	D7	D6	D5	D4	D3	D2	D1	D0	DATA
Second byte	1	0	R5	R4	R3	R2	R1	R0	REGISTER #

1.2 Indirect access

Specify the register number in control register R#17 (Control Register Pointer).

First set the register number in R#17 (using direct addressing) by sending data to port #3. When you set the data in R#17, you can also set its MSB (AII, the autoincrement bit) to control autoincrementing. The data in R#17 cannot be changed by indirect addressing.

If autoincrement is prohibited, the contents of R#17 will be unchanged, and thus you do not have to reset R#17.

	MSB 7	6	5	4	3	2	1	0 LSB	
Register #17	AII	0	R5	R4	R3	R2	R1	R0	REGISTER #
	1: Auto increment inhibit 0: Auto increment on								
Port #3 First byte	D7	D6	D5	D4	D3	D2	D1	D0	DATA
Port #3 Second byte	D7	D6	D5	D4	D3	D2	D1	D0	DATA
	...								
Port #3 nth byte	D7	D6	D5	D4	D3	D2	D1	D0	DATA

2. Accessing the Palette Registers

To set data in the MSX-VIDEO palette registers (P#0 to P#15/9 bit), you must first set the palette register number in register R#16 (Color palette address pointer) and subsequently output the two bytes of data (in order) through port #2.

	MSB 7	6	5	4	3	2	1	0 LSB	
Register #16	0	0	0	0	C3	C2	C1	C0	Palette #
Port #2 First byte	0	R2	R1	R0	0	B2	B1	B0	DATA
		Red data				Blue data			
Port #2 Second byte	0	0	0	0	0	G2	G1	G0	DATA
						Green data			

(Editor's note: R#16 increases after every second data byte written)

3. Accessing the Status Registers

To read the status registers of the MSX-VIDEO (S#0 to S#9), you must first set the register number in R#15 (Status register pointer) and read the data through Port #1.

	MSB 7	6	5	4	3	2	1	0 LSB	
Register #15	0	0	0	0	S3	S2	S1	S0	Status register
Port #1 Read data	D7	D6	D5	D4	D3	D2	D1	D0	DATA

4. Accessing the Video RAM

A Video RAM of 128K bytes plus an expansion RAM of 64K bytes can be connected to the MSX-VIDEO. The memory maps for these cases are shown in the map below.

```
                 Address counter

  +---------+        1FFFFH
  |         |           |
  |         |           |
  |         |           |
  |         |           |
  +---------+        0FFFFH        +---------+
  |         |           |          |         |
  |         |           |          |         |
  |         |           |          |         |
  |         |           |          |         |
  +---------+        00000H        +---------+

   Video RAM                       Expansion RAM
 (For display)                (For data I/O registers)
```

Accessing memory

To access memory, follow the procedure below.

1. Switch banks (VRAM to Expansion RAM)
2. Set the address counter (A16 to A14)
3. Set the address counter (A7 to A0)
4. Set the address counter (A13 to A8), and specify read or write
5. Read or write the data

1. Switching banks (VRAM to Expansion RAM)

Since the contents of R#45 (Argument register) do no change each time that memory is accessed, it is not necessary to respecify bit 6 of register R#45 (which specifies banking) every time that you are to do banking.

	MSB	7	6	5	4	3	2	1	0	LSB
Register #45		0	MXC	MXD	MXS	DIY	DIX	EQ	MAJ	Argument register

MXC: 1: Expansion RAM
0: Video RAM

2. Setting the address counter (A16-A14)

Set the high-order three bits (A16 to A14) of the address counter using register R#14 (VRAM Access base address register).

	MSB	7	6	5	4	3	2	1	0	LSB
Register #14		0	0	0	0	0	A16	A15	A14	VRAM Access base address register

3. Setting the address counter (A7 to A0)

Set the low-order eight bits (A7 to A0) of the address counter by outputting data to Port #1.

	MSB	7	6	5	4	3	2	1	0	LSB
Port #1		A7	A6	A5	A4	A3	A2	A1	A0	First byte

4. Setting the address counter (A13 to A8) and specifying read or write

Set the remaining six bits (A13 to A8) of the address counter and specify read or write by outputting data to Port #1.

	MSB	7	6	5	4	3	2	1	0	LSB
Port #1		X	X	A13	A12	A11	A10	A9	A8	Second byte

0 0: Read
0 1: Write

5. Reading or writing data

Since the address counter is automatically incremented when data is read from or written to Port #0, you may continually access blocks of data.

- To access the VRAM, you can also use commands. These commands will be explained in a later chapter.
- Refer to the data sheet for access timings.

REGISTER FUNCTIONS

1. CONTROL REGISTERS

#0 to #23 (Write only)
#32 to #46 (Write only)

1.1 Mode Registers

	MSB 7	6	5	4	3	2	1	0 LSB	
R#0	0	DG	IE2	IE1	M5	M4	M3	0	Mode Register 0
R#1	0	BL	IE0	M1	M2	0	SI	MAG	Mode Register 1
R#8	MS	LP	TP	CB	VR	0	SPD	BW	Mode Register 2
R#9	LN	0	S1	S0	IL	EO	$\overline{NT}$	DC	Mode Register 3

R#0 DG: Sets the color bus to input mode, and inputs data into the VRAM.
IE2: Enables interrupt from Lightpen by Interrupt Enable 2.
IE1: Enables interrupt from Horizontal scanning line by Interrupt Enable 1.
M5: Used to change the display mode.
M4: Used to change the display mode.
M3: Used to change the display mode.

R#1 BL: When 1, screen display enabled. When 0, screen disabled.
IE0: Enables interrupt from Horizontal scanning line by Interrupt Enable 0.
M1: Used to change the display mode.
M2: Used to change the display mode.
SI: When 1, sprite size is 16 x 16. When 0, 8 x 8.
MA: Sprite expansion; when 1: expanded. When 0, normal.

R#8 MS: When 1, sets the color bus to input mode and enables mouse.
When 0, sets the color bus to output mode and disables mouse.
LP: When 1, enables light pen. When 0, disables light pen.
TP: Sets the color of code 0 to the color of the palette.
CB: When 1, sets the color bus to input mode.
When 0, sets the color bus to output mode.
VR: Selects the type of Video RAM.
1 = 64K x 1 bit or 64K x 4 bits.
0 = 16K x 1 bit or 16K x 4 bits.
SPD: When 1, disables display of sprites. When 0, displays sprites.
BW: When 1, sets black and white in 32 tones.
When 0, sets color (available only with a composite encoder).

R#9	LN:	When 1, sets the horizontal dot count to 212. When 0, sets the horizontal dot count to 192.
	S1:	Selects simultaneous mode.
	S0:	Selects simultaneous mode.
	IL:	When 1, interlace (Complete NTSC timing) When 0, non-interlace (Incomplete NTSC timing)
	EO:	When 1, displays two graphic screens interchangably by Even field/Odd field. When 0, displays the same graphic screen by Even field/Odd field.
	$\overline{\text{NT}}$:	When 1, PAL (313 lines); when 0, NTSC (262 lines). (For RGB output only)
	DC:	When 1, sets *DLCLK to input mode; when 0, sets *DLCLK to output mode.

1.2 Table Base Address Registers

The table base address registers are a set of registers to declare the addresses of tables in the VRAM to be used by MSX-VIDEO.

Note that when these registers are accessed, the control codes that the screen may receive depends on the display mode. For this purpose, you must mask the unwanted bits.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#2     | 0 |A16|A15|A14|A13|A12|A11|A10| Pattern name table base
           +---+---+---+---+---+---+---+---+ address register
           +---+---+---+---+---+---+---+---+
   R#3     |A13|A12|A11|A10|A9 |A8 |A7 |A6 | Color table base address
           +---+---+---+---+---+---+---+---+ register low
           +---+---+---+---+---+---+---+---+
   R#10    | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| Color table base address
           +---+---+---+---+---+---+---+---+ register high
           +---+---+---+---+---+---+---+---+
   R#4     | 0 | 0 |A16|A15|A14|A13|A12|A11| Pattern generator table
           +---+---+---+---+---+---+---+---+ base address register
           +---+---+---+---+---+---+---+---+
   R#5     |A14|A13|A12|A11|A10|A9 |A8 |A7 | Sprite attribute table
           +---+---+---+---+---+---+---+---+ base address register low
           +---+---+---+---+---+---+---+---+
   R#11    | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15| Sprite attribute table
           +---+---+---+---+---+---+---+---+ base address register high
           +---+---+---+---+---+---+---+---+
   R#6     | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
           +---+---+---+---+---+---+---+---+ table base address register
```

1.3 Color Registers

The color registers are used to control the MSX-VIDEO's text and background screen colors as well as blinking, etc.

```
     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
         +---+---+---+---+---+---+---+---+ color register

           TC3 to TC0:  Specifies the text color according to TEXT 1
                        and TEXT 2 modes.
           BD3 to BD0:  Specifies the back drop color in all display
                        modes.
     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#12    |T23|T22|T21|T20|BC3|BC2|BC1|BC0| Text color/Back color
         +---+---+---+---+---+---+---+---+ register
```

In TEXT 2 mode, if the attributes for blinking are set, the color set in this register and set in R#7 are displayed alternately.

```
           T23 to T20:  Specifies the color of part 1 of the pattern.
           BC3 to BC0:  Specifies the color of part 0 of the pattern.

     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#13    |ON3|ON2|ON1|ON0|OF3|OF2|OF1|OF0| Blinking period register
         +---+---+---+---+---+---+---+---+
```

In the bit map modes of GRAPH4 to GRAPH7, the two pages are alternately displayed (blinked). Place data in this register to set the display page to an odd page to begin blinking. This register is also used in the TEXT2 mode.

```
           ON3 to ON0:  Display time for even page
           OF3 to OF0:  Display time for odd page

     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#20    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Color burst register 1
         +===+===+===+===+===+===+===+===+
 R#21    | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Color burst register 2
         +===+===+===+===+===+===+===+===+
 R#22    | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | Color burst register 3
         +---+---+---+---+---+---+---+---+
```

The above values are preset when the power is applied. If all values in the above three registers are set to 0, the color burst signal of the composite video output will be erased.

If the above values are subsequently reset to the preset values, the normal color burst signal will be output.

1.4 Display Registers

The display registers are used to control the display position on the CRT.

```
    MSB  7   6   5   4   3   2   1   0  LSB
       +---+---+---+---+---+---+---+---+
R#18   |V3 |V2 |V1 |V0 |H3 |H2 |H1 |H0 | Display adjust register
       +---+---+---+---+---+---+---+---+
```

The above register is used to adjust the display position on the CRT.

```
H = 7 . . . H = 1,  H = 0, H = 15 . . . H = 8
(Left)            (Center)             (Right)

V = 8 . . . V = 15, V = 0, V = 1  . . . V = 7
(Bottom)          (Center)             (Top)
```

(Editor's note: modifying this value while executing a VDP command will result in a corrupted byte)

```
    MSB  7   6   5   4   3   2   1   0  LSB
       +---+---+---+---+---+---+---+---+
R#23   |DO7|DO6|DO5|DO4|DO3|DO2|DO1|DO0| Display offset register
       +---+---+---+---+---+---+---+---+
```

The above register sets the location of the line to begin display.

```
                VRAM (Line)

       TOP  +------------+ 0
            |            |
            | CRT Screen |
            |            |
     BOTTOM +------------+ 192 or 212   R#23 = 0
            |            |
            +------------+ 255

            +------------+ 0
            |            |
            | CRT Screen |
            |            |
     BOTTOM +------------+ 136 or 156   R#23 = 200
            |            |
       TOP  +------------+ 200
            | CRT Screen |
            +------------+ 255
    MSB  7   6   5   4   3   2   1   0  LSB
       +---+---+---+---+---+---+---+---+
R#19   |IL7|IL6|IL5|IL4|IL3|IL2|IL1|IL0| Interrupt line register
       +---+---+---+---+---+---+---+---+
```

You may specify interrupts when the MSX-VIDEO begins to display a specified scanning line. To enable the interrupt, use the above register to set the scanning line.

1.5 ACCESS REGISTERS

The access registers are a set of registers used when accessing the MSX-VIDEO registers or the VRAM.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#14    | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| VRAM Access base address
           +---+---+---+---+---+---+---+---+ register
```

When accessing the MSX-VIDEO and the Video RAM (VRAM), set the high-order three bits of the address in the VRAM access base address register.

When data is set in this register, and the VRAM is accessed, if there is a carry from A13, the data in the register is automatically incremented. In GRAPHIC1, GRAPHIC2, MULTICOLOR, and TEXT1 modes, the data in the register is not automatically incremented.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#15    | 0 | 0 | 0 | 0 |S3 |S2 |S1 |S0 | Status register pointer
           +---+---+---+---+---+---+---+---+
```

When reading the MSX-VIDEO status registers (S#0 to S#9), set the contents of the status register pointer.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#16    | 0 | 0 | 0 | 0 |C3 |C2 |C1 |C0 | Color palette address
           +---+---+---+---+---+---+---+---+ register
```

When setting the color palette of the MSX-VIDEO, set the number of the palette in the Color palette address register.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#17    |AII| 0 |RS5|RS4|RS3|RS2|RS1|RS0| Control register pointer
           +---+---+---+---+---+---+---+---+
```

In the MSX-VIDEO, the above control register pointer may be used to access another register. In addition, according to the setting of the AII bit, the data van be automatically incremented.

```
    AII = 1:  Auto increment disabled
    AII = 0:  Auto increment enabled
```

1.6 COMMAND REGISTERS

The following command registers are used when executing a command on the MSX-VIDEO. Details on the use of these command registers will be presented in a later chapter.

```
  MSB  7   6   5   4   3   2   1   0  LSB
     +---+---+---+---+---+---+---+---+
R#32 |SX7|SX6|SX5|SX4|SX3|SX2|SX1|SX0| Source X low register
     +===+===+===+===+===+===+===+===+
R#33 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |SX8| Source X high register
     +===+===+===+===+===+===+===+===+
R#34 |SY7|SY6|SY5|SY4|SY3|SY2|SY1|SY0| Source Y low register
     +===+===+===+===+===+===+===+===+
R#35 | 0 | 0 | 0 | 0 | 0 | 0 |SY9|SY8| Source Y high register
     +---+---+---+---+---+---+---+---+
     +---+---+---+---+---+---+---+---+
R#36 |DX7|DX6|DX5|DX4|DX3|DX2|DX1|DX0| Destination X low register
     +===+===+===+===+===+===+===+===+
R#37 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |DX8| Destination X high register
     +===+===+===+===+===+===+===+===+
R#38 |DY7|DY6|DY5|DY4|DY3|DY2|DY1|DY0| Destination Y low register
     +===+===+===+===+===+===+===+===+
R#39 | 0 | 0 | 0 | 0 | 0 | 0 |DY9|DY8| Destination Y high register
     +---+---+---+---+---+---+---+---+
     +---+---+---+---+---+---+---+---+
R#40 |NX7|NX6|NX5|NX4|NX3|NX2|NX1|NX0| Number of dots X low register
     +===+===+===+===+===+===+===+===+
R#41 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |NX8| Number of dots X high register
     +===+===+===+===+===+===+===+===+
R#42 |NY7|NY6|NY5|NY4|NY3|NY2|NY1|NY0| Number of dots Y low register
     +===+===+===+===+===+===+===+===+
R#43 | 0 | 0 | 0 | 0 | 0 | 0 |NY9|NY8| Number of dots Y high register
     +---+---+---+---+---+---+---+---+
     +---+---+---+---+---+---+---+---+
R#44 |CH3|CH2|CH1|CH0|CL3|CL2|CL1|CL0| Color register
     +===+===+===+===+===+===+===+===+
R#45 | 0 |MXC|MXD|MXS|DIY|DIX|EQ |MAJ| Argument register
     +===+===+===+===+===+===+===+===+
R#46 |CM3|CM2|CM1|CM0|LO3|LO2|LO1|LO0| Command register
     +---+---+---+---+---+---+---+---+
```

2. STATUS REGISTERS #0 to #9 (Read only)

The following status registers are read-only registers for reporting the status when the MSX-VIDEO is read.

```
   MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
S#0     | F |5S | C |Fifth sprite number| Status register 0
        +---+---+---+---+---+---+---+---+
  F:    Vertical scanning interrupt flag
        When S#0 is read, this flag is reset.

  5S:   Flag for the fifth sprite
        Five sprites are aligned on the first horizontal line (In
        the G3 to G7 modes, 9 sprites are allowed)

  C:    Collision flag
        Two sprites have collided.

  Fifth sprite number:
        The number of the fifth (or ninth) sprite.

   MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
S#1     |FL |LPS| Identification #  |FH | Status register 1
        +---+---+---+---+---+---+---+---+
  FL:   Lightpen flag (Lightpen flag set)
        If the lightpen is to detect light, this bit as well as the
        IE2 bit must both set in order for an interrupt to be
        enabled. When S#1 is read, FL is reset.

        Mouse switch 2 (Mouse flag set)
        The second switch on the mouse was pressed.
        In this case, when S#1 is read, FL is not reset.

  LPS:  Lightpen switch (Lightpen flag set)
        The lightpen switch was pressed.
        In this case, when S#1 is read, LPS is not reset.

        Mouse switch 1 (Mouse flag set)
        The first switch on the mouse was pressed.
        In this case, when S#1 is read, LPS is not reset.

  Identification number:
        The identification number (ID #) of the MSX-VIDEO.
        (Editor´s note: 0 if V9938, 1 if V9948, 2 if V9958)

  FH:   Horizontal scanning interrupt flag
        Horizontal scanning interrupt (which is specified in R#19)
        flag. If IE1 is set, an interrupt is enabled. When S#1
        is read, FH is reset.

   MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
S#2     |TR |VR |HR |BD | 1 | 1 |EO |CE | Status register 2
        +---+---+---+---+---+---+---+---+
  TR:   Transfer ready flag
        When the CPU sends commands to the VRAM and other devices,
        the CPU checks this flag while transferring data. When this
        flag is set to 1, transfer may be done.
```

```
VR:  Vertical scanning line timing flag
     During vertical scanning, this flag is set to 1.

HR:  Horizontal scanning line timing flag
     During horizontal scanning, this flag is set to 1.

BD:  Boundary color detect flag
     When the search command is executed, this flag detects
     whether the boundary color was detected or not.

EO:  Display field flag
     When 0, indicates the first field.
     When 1, indicates the second field.

CE:  Command execution flag
     Indicates that a command is being executed.
```

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   S#3     |X7 |X6 |X5 |X4 |X3 |X2 |X1 |X0 | Column register low
           +===+===+===+===+===+===+===+===+
   S#4     | 1 | 1 | 1 | 1 | 1 | 1 | 1 |X8 | Column register high
           +===+===+===+===+===+===+===+===+
   S#5     |Y7 |Y6 |Y5 |Y4 |Y3 |Y2 |Y1 |Y0 | Row register low
           +===+===+===+===+===+===+===+===+
   S#6     | 1 | 1 | 1 | 1 | 1 | 1 |EO |Y8 | Row register high
           +---+---+---+---+---+---+---+---+
```

The above registers are set to indicate the collision location of sprites, the location of lightpen detection, and the relative movement of the mouse.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   S#7     |C7 |C6 |C5 |C4 |C3 |C2 |C1 |C0 | Color register
           +---+---+---+---+---+---+---+---+
```

The above color register is used when the POINT and VRAM to CPU commands are executed. The VRAM data is set in this register.

```
       MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   S#8     |BX7|BX6|BX5|BX4|BX3|BX2|BX1|BX0| Border X register low
           +===+===+===+===+===+===+===+===+
   S#9     | 1 | 1 | 1 | 1 | 1 | 1 | 1 |BX8| Border X register high
           +---+---+---+---+---+---+---+---+
```

When the search command is executed and the border color has been detected, the X coordinate is set in the above registers.

TEXT 1 MODE

Characteristics

```
- Pattern size              : 6 dots (w) x 8 dots (h)
- Patterns                  : 256 types
- Screen pattern count      : 40 (w) x 24 (h) patterns
- Pattern colors            : Two colors out of 512 colors (per screen)
- VRAM area per screen      : 4K bytes
```

Controls

```
- Pattern font              : VRAM pattern generator table
- Screen pattern location   : VRAM pattern name table
- Pattern color code 1      : High-order four bits of R#7
- Pattern color code 0      : Low-order fout bits of R#7
- Background color code     : Low-order fout bits of R#7
```

Initial settings

1. Mode and register settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 0*| 0*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 1*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in TEXT 1 mode
** Indicates negative logic
All other bits are set accordingly
```

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern had a number from PN0 to PN255.
- The font for each pattern is constructed from 8 bytes, and the lower two bits of each of the eight bytes is not displayed.
- Set the beginning (head) address of the pattern generator table is register R#4.

```
     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
R#4      | 0 | 0 |A16|A15|A14|A13|A12|A11| Pattern generator table
         +---+---+---+---+---+---+---+---+ base address register
```

Pattern generator table

```
(X=1, O=0)

               ++--------  These bits are not displayed.
               ||
     MSB 76543210 LSB
       0 OOXOOOOO -----+
       1 OXOXOOOO      |
       2 XOOOXOOO      |
       3 XOOOXOOO      +--- Pattern number 0
       4 XXXXXOOO      |
       5 XOOOXOOO      |
       6 XOOOXOOO      |
       7 OOOOOOOO -----+
       8 XXXXOOOO -----+
       9 XOOOXOOO      |
      10 XOOOXOOO      |
      11 XXXXOOOO      +--- Pattern number 1
      12 XOOOXOOO      |
      13 XOOOXOOO      |
      14 XXXXOOOO      |
      15 OOOOOOOO -----+
       .      .      .
       .      .      .
       .      .      .
    2040 XOXOXOOO -----+
    2041 OXOXOXOO      |
    2042 XOXOXOOO      |
    2043 OXOXOXOO      +--- Pattern number 0
    2044 XOXOXOOO      |
    2045 OXOXOXOO      |
    2046 XOXOXOOO      |
    2047 OXOXOXOO -----+
```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- Set the beginning (head) address of the pattern name table in register R#2.

```
     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
R#2      | 0 |A16|A15|A14|A13|A12|A11|A10| Pattern name table
         +---+---+---+---+---+---+---+---+ base address register
```

Pattern name table

```
|         |
+---------+ Base address
| ( 0, 0) |   0            0    1    2    3   .  .   39  X
+---------+                 +----+----+----+----+      +----+
| ( 1, 0) |   1          0 |    0|    1|    2|    3| .  . |  39|
+---------+                 +----+----+----+----+      +----+
| ( 2, 0) |   2          1 |  40|  41|  42|  43| .  . |  79|
+---------+                 +----+----+----+----+      +----+
.         .   .          .    .    .    .    .   .  .    .
.         .   .          .    .    .    .    .   .  .    .
.         .   .             +----+----+              --+----+
+---------+             22 | 880| 881|  .    .   .  . | 919|
| (39, 0) |  39             +----+----+              --+----+
+---------+             23 | 920| 921|  .    .   .  . | 959|
| ( 0, 1) |  40             +----+----+              --+----+
+---------+              Y
.         .   .
.         .   .          Screen display correspondence
.         .   .
+---------+
| (39,23) | 959
+---------+
```

4. Color register settings

```
      MSB  7   6   5   4   3   2   1   0  LSB
          +---+---+---+---+---+---+---+---+
  R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop color
          +--++--++--++--++--++--++--++--++ register
             |   |   |   |   +---+---+---+-- Specifies pattern color
             |   |   |   |                   code 0 or back drop color
             +---+---+---+------------------ Specifies pattern color
                                             code 1
```

Example of VRAM allocation in TEXT 1 mode

```
00000H  +-----------+         MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |             +---+---+---+---+---+---+---+---+
        | Name      |        R#2  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
        | Table 0   |             +---+---+---+---+---+---+---+---+
        |           |                A16 A15 A14 A13 A12 A11 A10
003C0H  +-----------+
        |           |
00800H  +-----------+         MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |             +---+---+---+---+---+---+---+---+
        | Generator|         R#4  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
        | Table 0   |             +---+---+---+---+---+---+---+---+
        |           |                    A16 A15 A14 A13 A12 A11
        |           |
01000H  +-----------+         MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |             +---+---+---+---+---+---+---+---+
        | Name      |        R#2  | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
        | Table 1   |             +---+---+---+---+---+---+---+---+
        |           |                A16 A15 A14 A13 A12 A11 A10
        |           |
013C0H  +-----------+
        |           |
        |           |
01800H  +-----------+         MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |             +---+---+---+---+---+---+---+---+
        | Generator|         R#4  | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
        | Table 1   |             +---+---+---+---+---+---+---+---+
        |           |                    A16 A15 A14 A13 A12 A11
02000H  +-----------+
        .           .
        .           .        A maximum of 32 pages may be allocated in
        .           .        the same manner (using a 128K-byte VRAM).
        .           .
        |           |
        |           |
1FFFFH  +-----------+
```

TEXT 2 MODE

Characteristics

```
- Pattern size            : 6 dots (w) x 8 dots (h)
- Patterns                : 256 types
- Screen pattern count    : 80 (w) x 24 (h) patterns
                            80 (w) x 26.5 (h) patterns
- Pattern blinking        : Possible for each character
- Pattern colors          : Two colors out of 512 colors (per screen),
                            four if using blinking
- VRAM area per screen    : 8K bytes
```

Controls

```
- Pattern font              : VRAM pattern generator table
- Screen pattern location   : VRAM pattern name table
- Blink attributes          : VRAM color table
- Pattern color code 1      : High-order four bits of R#7
- Pattern color code 0      : Low-order four bits of R#7
- Background color code     : Low-order four bits of R#7
- Pattern color code 1      : High-order four bits of R#12
                              (used for blinking)
- Pattern color code 0      : Low-order four bits of R#12
                              (used for blinking)
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 1*| 0*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 1*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in TEXT 2 mode
** Indicates negative logic
```

In this display mode, if LN is set to 1, 26.5 lines are selected, and if LN is set to 0, 24 lines are selected.
All other bits are set accordingly.

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255.
- Set the beginning (head) address of the pattern generator table in register R#4

```
      MSB   7   6   5   4   3   2   1   0   LSB
           +---+---+---+---+---+---+---+---+
R#4        | 0 | 0 |A16|A15|A14|A13|A12|A11| Pattern generator table
           +---+---+---+---+---+---+---+---+ base address register
```

- The font for each pattern is constructed from 8 bytes, and the lower two bits of each of the eight bytes is not displayed.

Pattern generator table

```
(X=1, O=0)

                 ++--------  These bits are not displayed.
                 ||
       MSB 76543210 LSB
         0 OOXOOOOO -----+
         1 OXOXOOOO      |
         2 XOOOXOOO      |
         3 XOOOXOOO      +--- Pattern number 0
         4 XXXXXOOO      |
         5 XOOOXOOO      |
         6 XOOOXOOO      |
         7 OOOOOOOO -----+
         8 XXXXOOOO -----+
         9 XOOOXOOO      |
        10 XOOOXOOO      |
        11 XXXXOOOO      +--- Pattern number 1
        12 XOOOXOOO      |
        13 XOOOXOOO      |
        14 XXXXOOOO      |
        15 OOOOOOOO -----+
         .       .      .
         .       .      .
         .       .      .
      2040 XOXOXOOO -----+
      2041 OXOXOXOO      |
      2042 XOXOXOOO      |
      2043 OXOXOXOO      +--- Pattern number 255
      2044 XOXOXOOO      |
      2045 OXOXOXOO      |
      2046 XOXOXOOO      |
      2047 OXOXOXOO -----+
```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- If LN is set to 0, the screen display pattern is 80 (W) x 24 (H); and if LN is set to 1, the screen display pattern is 80 (W) x 26.5 (H). The upper half of the 27th pattern is displayed.
- Set the beginning (head) address of the pattern name table in register R#2.

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
 R#2       | 0 |A16|A15|A14|A13|A12| 1 | 1 | Pattern name table
           +---+---+---+---+---+---+---+---+  base address register
```

Pattern name table

```
  |         |
  +---------+ Base address
  | ( 0, 0) |   0            0    1    2    3    .  .   79  X
  +---------+                +----+----+----+----+      +----+
  | ( 1, 0) |   1          0 |   0|   1|   2|   3| .  . |  79|
  +---------+                +----+----+----+----+      +----+
  | ( 2, 0) |   2          1 |  80|  81|  82|  83| .  . | 159|
  +---------+                +----+----+----+----+      +----+
  .         .   .          .    .    .    .    .    .  .    .
  .         .   .          .    .    .    .    .    .  .    .
  .         .   .            +----+----+              --+----+
  +---------+             25 |2000|2001|  .    .    .  . |2079|
  | (79, 0) |  79            +----+----+              --+----+
  +---------+             26 |2080|2081|  .    .    .  . |2159|
  | ( 0, 1) |  80            +----+----+              --+----+
  +---------+              Y
  .         .   .
  .         .   .          Screen display correspondence
  .         .   .
  +---------+
  | (79,26) | 2159
  +---------+
```

4. Color table settings

- In TEXT 2 mode, each pattern has a separate bit for the attribute area, and if this bit is set to 1, the pattern blink attribute will be set.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
 R#3       |A13|A12|A11|A10|A9 | 1 | 1 | 1 | Color table
           +===+===+===+===+===+===+===+===+
 R#10      | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| base address registers
           +---+---+---+---+---+---+---+---+
```

Color table

```
 MSB   7       6       5       4       3       2       1       0   LSB
   +-------+-------+-------+-------+-------+-------+-------+-------+
  0|( 0, 0}|( 1, 0}|( 2, 0}|( 3, 0}|( 4, 0}|( 5, 0}|( 6, 0}|( 7, 0}| Base
   +-------+-------+-------+-------+-------+-------+-------+-------+ address
  1|( 8, 0}|( 9, 0}|(10, 0}|(11, 0}|(12, 0}|(13, 0}|(14, 0}|(15, 0}|
   +-------+-------+-------+-------+-------+-------+-------+-------+
  .    .       .       .       .       .       .       .       .
  .    .       .       .       .       .       .       .       .
  .    .       .       .       .       .       .       .       .
  .    .       .       .       .       .       .       .       .
   +-------+-------+-------+-------+-------+-------+-------+-------+
269|(72,26}|   .   |   .   |   .   |   .   |   .   |   .   |(79,26}|
   +-------+-------+-------+-------+-------+-------+-------+-------+
```

5. Color register settings

- Set the color for pattern 1 in the high-order bits of register R#7.
- Set the color for pattern 0 in the low-order bits of register R#7.

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
           +---+---+---+---+---+---+---+---+ color register
```

- Set the blink attribute for the corresponding pattern by setting an alternate color code in register R#12. The pattern will be blinked by using the color codes set in register R#7 and R#12.

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#12    |T23|T22|T21|T20|BC3|BC2|BC1|BC0| Text color/Back color
           +---+---+---+---+---+---+---+---+ register
```

6. Blink register settings

- The color codes set in registers R#7 and R#12 will be alternately displayed for blinking; however, the blinking period attribute (time on and time off) can also be set in register R#13.

```
                 +-----------------+                +-------------
                 |                 |                |
                 |      R#12       |      R#7       |      R#12
                 |                 |                |
        ---------+                 +----------------+

                 |<-- On time --->|<-- Off time -->|
```

```
     MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
R#13     |ON3|ON2|ON1|ON0|OF3|OF2|OF1|OF0| Blinking period register
         +--++---+---+--++--++---+---+--++
            +-----------+   +-----------+
               ON time         OFF time
```

- The data for the ON and OFF times are shown below. (NTSC)

DATA (binary)	TIME (ms)	VBlanks
0 0 0 0	0	0
0 0 0 1	166.9	10
0 0 1 0	333.8	20
0 0 1 1	500.5	30
0 1 0 0	667.5	40
0 1 0 1	843.4	50
0 1 1 0	1001.3	60
0 1 1 1	1168.2	70
1 0 0 0	1335.1	80
1 0 0 1	1501.9	90
1 0 1 0	1668.8	100
1 0 1 1	1835.7	110
1 1 0 0	2002.6	120
1 1 0 1	2169.5	130
1 1 1 0	2336.3	140
1 1 1 1	2503.2	150

Example of VRAM allocation in TEXT 2 mode

```
                                  Pattern name table base address

00000H  +-----------+              MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |                  +---+---+---+---+---+---+---+---+
        | Name      |             R#2  | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
        | Table 0   |                  +---+---+---+---+---+---+---+---+
00870H  +-----------+                       A16 A15 A14 A13 A12
        |           |
00A00H  + Color    -+
        | Table 0   |             Color table base address
00B0EH  +-----------+
00800H  +-----------+              MSB   7   6   5   4   3   2   1   0   LSB
        | Pattern   |                  +---+---+---+---+---+---+---+---+
        | Generator|              R#3  | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
        | Table 0   |                  +---+---+---+---+---+---+---+---+
01800H  +-----------+                   A13 A12 A11 A10 A9
        | Pattern   |                  +---+---+---+---+---+---+---+---+
        | Generator|              R#10 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
        | Table 0   |                  +---+---+---+---+---+---+---+---+
02000H  +-----------+                                       A16 A15 A14
        | Pattern   |
        | Name      |
        | Table 1   |             Pattern generator table base address
02870H  +-----------+
02A00H  + Color    -+              MSB   7   6   5   4   3   2   1   0   LSB
        | Table 1   |                  +---+---+---+---+---+---+---+---+
02B0EH  +-----------+             R#4  | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
03000H  +-----------+                  +---+---+---+---+---+---+---+---+
        | Pattern   |                           A16 A15 A14 A13 A12 A11
        | Generator|
        | Table 0   |
        |           |
03800H  +-----------+
        |           |
04000H  +-----------+
        .           .
        .           .             A maximum of 16 pages may be allocated in
        .           .             the same manner (using a 128K-byte VRAM).
        .           .
        |           |
        |           |
1FFFFH  +-----------+
```

MULTICOLOR MODE

Characteristics

```
- Screen composition          : 64 (w) x 48 (h) color blocks
- Color blocks                : Sixteen colors out of 512 colors
- Sprite mode                 : Sprite mode 1
- VRAM area per screen        : 4K bytes
```

Controls

```
- Color block color code      : VRAM pattern generator table
- Color block location        : VRAM pattern name table
- Background color code       : Low-order four bits of R#7
- Sprites                     : VRAM sprite attribute table
                                VRAM sprite pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 0*| 0*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 1*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in MULTICOLOR mode
** Indicates negative logic
```

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the colors of the color blocks.
- Each pattern is made up of four color blocks. These patterns are approximately 8 x 8 when the dots available for the screen display area is 256 x 192 dots.

```
<----  8 dots  --->

+--------+--------+    ^
|        |        |    |
|   A    |   B    |    |
|        |        |    |        For each block A, B, C,
+--------+--------+ 8 dots      and D, sixteen colors may
|        |        |    |        be specified.
|   C    |   D    |    |
|        |        |    |
+--------+--------+    v
```

- In the MULTICOLOR mode, two bytes are used for each pattern, and each pattern includes four color blocks.

```
MSB  7   6   5   4   3   2   1   0   LSB
   +---+---+---+---+---+---+---+---+
   |  Color code A |  Color code B |
   +===+===+===+===+===+===+===+===+
   |  Color code C |  Color code D |
   +---+---+---+---+---+---+---+---+
```

- In the MULTICOLOR mode, for each pattern name, there are four corresponding color blocks, and according to the y-coordinate, the pattern names are automatically set.

```
              MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+ +---+---+---+---+---+---+---+---+
  +-    | A | B | |  Color code A |  Color code B | Pattern when
  |     +---+---+ +---+---+---+---+---+---+---+---+ Y=0, 4, 8, 12, 16
  |     | C | D | |  Color code C |  Color code D | or 20 is selected
  |     +---+---+ +---+---+---+---+---+---+---+---+
  |     +---+---+ +---+---+---+---+---+---+---+---+
  |     | E | F | |  Color code E |  Color code F | Pattern when
  |     +---+---+ +---+---+---+---+---+---+---+---+ Y=1, 5, 9, 13, 17
Pattern | G | H | |  Color code G |  Color code H | or 21 is selected
 name   +---+---+ +---+---+---+---+---+---+---+---+
  #N    +---+---+ +---+---+---+---+---+---+---+---+
  |     | I | J | |  Color code I |  Color code J | Pattern when
  |     +---+---+ +---+---+---+---+---+---+---+---+ Y=2, 6, 10, 14, 18
  |     | K | L | |  Color code K |  Color code L | or 22 is selected
  |     +---+---+ +---+---+---+---+---+---+---+---+
  |     +---+---+ +---+---+---+---+---+---+---+---+
  |     | M | N | |  Color code M |  Color code N | Pattern when
  |     +---+---+ +---+---+---+---+---+---+---+---+ Y=3, 7, 11, 15, 19
  +-    | O | P | |  Color code O |  Color code P | or 23 is selected
        +---+---+ +---+---+---+---+---+---+---+---+
```

- Set the beginning (head) address of the pattern generator table in register R#4.

```
      MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
  R#4    | 0 | 0 |A16|A15|A14|A13|A12|A11| Pattern generator table
         +---+---+---+---+---+---+---+---+  base address register
```

Pattern generator table

```
|                  |
+------------------+     0  Base address
| Pattern name #0  |
|   (Eight bytes)  |
+------------------+     8
| Pattern name #1  |
|   (Eight bytes)  |
+------------------+    16
.                  .
.                  .
.                  .
+------------------+  2040
| Pattern name #255 |
|   (Eight bytes)  |
+------------------+  2048
```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern number.

Pattern name table

	0	**1**	**2**	**3**	**...**	**31**
0	0	1	2	3	...	31
1	32	33	34	35	...	63
...	...	...	...	...	...	...
...	...	...	...	...	...	...
22	704	705	...	...	...	735
23	736	737	...	...	...	767

- Set the beginning (head) address of the pattern name table in register R#2.

	MSB 7	6	5	4	3	2	1	0 LSB	
R#2	0	A16	A15	A14	A13	A12	A11	A10	Pattern name table base address register

Pattern name table

```
|         |
+---------+ Base address
| ( 0, 0) |   0
+---------+
| ( 1, 0) |   1
+---------+
| ( 2, 0) |   2
+---------+
.         .   .
.         .   .
.         .   .
+---------+
| (31, 0) |  31
+---------+
| ( 0, 1) |  32
+---------+
.         .   .
.         .   .
.         .   .
+---------+
| (31,23) | 767
+---------+
```

4. Color register settings

```
    MSB  7   6   5   4   3   2   1   0   LSB
       +---+---+---+---+---+---+---+---+
R#7    |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop color
       +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ register
         |   |   |   |   |   |   |   |
         |   |   |   |   +---+---+---+-- Specifies backdrop color
         +---+---+---+------------------ Ignored
```

5. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 1.

```
    MSB  7   6   5   4   3   2   1   0   LSB
       +---+---+---+---+---+---+---+---+
R#5    |A14|A13|A12|A11|A10|A9 |A8 |A7 | Sprite attribute table
       +===+===+===+===+===+===+===+===+ base address register
R#11   | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15|
       +---+---+---+---+---+---+---+---+
       +---+---+---+---+---+---+---+---+
R#6    | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
       +---+---+---+---+---+---+---+---+  table base address
                                          register
```

Example of VRAM allocation in MULTICOLOR mode

```
00000H  +----------+
        | Sprite   |
        | Generator|
        | Table    |
        |  (1024   |
        |   bytes) |
00400H  +----------+
        | Pattern  |
        | Name     |    .+----------+ 00700H
        | Table    |   . | Sprite   |
        |  (768    |  .  | attribute|
        |   bytes) | .   | table    |
00700H  +----------+.    |  (128    |
        |          |     |   bytes) |
        |          |     +----------+ 00780H
00800H  +----------+     |          |
        |          |.    |          |
        |          | .   |          |
        |          |  .  |          |
        |          |   . |          |
        |          |    .+----------+ 00800H
        |          |
        | Pattern  |
00C00H  | Generator|
        | Table    |
        |  (2048   |
        |   bytes) |
        |          |
        |          |
        |          |
01000H  +----------+
        .          .
        .          .        A maximum of 32 pages may be allocated in
        .          .        the same manner (using a 128K-byte VRAM).
        .          .
        |          |
        |          |
1FFFFH  +----------+
```

GRAPHIC 1 MODE

Characteristics

```
- Pattern size               : 8 dots (w) x 8 dots (h)
- Patterns                   : 256 types
- Screen pattern count       : 32 (w) x 24 (h) patterns
- Pattern colors             : 16 colors out of 512 colors (per screen)
- Sprite mode                : Sprite mode 1
- VRAM area per screen       : 4K bytes
```

Controls

```
- Pattern font               : VRAM pattern generator table
- Screen pattern location    : VRAM pattern name table
- Pattern color codes 1 & 0: Can be specified as a group for each
                             8-pattern set, in the VRAM color table
- Background color code      : Low-order four bits of R#7
- Sprites                    : VRAM sprite attribute table, VRAM sprite
                             pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 0*| 0*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 1 mode
** Indicates negative logic
```

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255.
- The font for each pattern is constructed from 8 bytes.
- Set the beginning (head) address of the pattern generator table in register R#4.

```
      MSB  7   6   5   4   3   2   1   0  LSB
          +---+---+---+---+---+---+---+---+
   R#4    | 0 | 0 |A16|A15|A14|A13|A12|A11| Pattern generator table
          +---+---+---+---+---+---+---+---+ base address register
```

Pattern generator table

```
(X=1, O=0)

     MSB 76543210 LSB
       0 OOXXXOOO -----+
       1 OXOOOXOO      |
       2 XOOOOOXO      |
       3 XOOOOOXO      +--- Pattern number 0
       4 XXXXXXXO      |
       5 XOOOOOXO      |
       6 XOOOOOXO      |
       7 OOOOOOOO -----+
       8 XXXXXXOO -----+
       9 XOOOOOXO      |
      10 XOOOOOXO      |
      11 XXXXXXOO      +--- Pattern number 1
      12 XOOOOOXO      |
      13 XOOOOOXO      |
      14 XXXXXXOO      |
      15 OOOOOOOO -----+
       .      .      .
       .      .      .
       .      .      .
    2040 XOXOXOXO -----+
    2041 OXOXOXOX      |
    2042 XOXOXOXO      |
    2043 OXOXOXOX      +--- Pattern number 255
    2044 XOXOXOXO      |
    2045 OXOXOXOX      |
    2046 XOXOXOXO      |
    2047 OXOXOXOX -----+
```

3. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- Set the beginning (head) address of the pattern name table in register R#2.

```
      MSB  7   6   5   4   3   2   1   0   LSB
          +---+---+---+---+---+---+---+---+
   R#2    | 0 |A16|A15|A14|A13|A12|A11|A10| Pattern name table
          +---+---+---+---+---+---+---+---+ base address register
```

Pattern name table

```
|         |
+---------+ Base address
| ( 0, 0) |   0            0    1    2    3   .  .   31 X
+---------+                +----+----+----+----+      +----+
| ( 1, 0) |   1          0 |   0|   1|   2|   3| .  . |  31
+---------+                +----+----+----+----+      +----+
| ( 2, 0) |   2          1 |  32|  33|  34|  35| .  . |  63
+---------+                +----+----+----+----+      +----+
.         .   .          .    .    .    .    .   .  .    .
.         .   .          .    .    .    .    .   .  .    .
.         .   .            +----+----+              --+----+
+---------+             22 | 704| 705|  .    .    .  . | 735|
| (31, 0) |  31            +----+----+              --+----+
+---------+             23 | 736| 737|  .    .    .  . | 767|
| ( 0, 1) |  32            +----+----+              --+----+
+---------+              Y
.         .   .
.         .   .          Screen display correspondence
.         .   .
+---------+
| (31,21) | 767 (byte)
+---------+
```

4. Color register settings

```
     MSB  7   6   5   4   3   2   1   0   LSB
        +---+---+---+---+---+---+---+---+
R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop color
        +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ register
          |   |   |   |   |   |   |   |
          |   |   |   |   +---+---+---+-- Specifies backdrop color
          +---+---+---+------------------ Ignored
```

5. Color table settings

- The colors for pattern color 1 and pattern color 0 are set in groups of eight patterns.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

```
     MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
R#3     |A13|A12|A11|A10|A9 |A8 |A7 |A6 | Color table
        +===+===+===+===+===+===+===+===+
R#10    | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| base address register
        +---+---+---+---+---+---+---+---+
```

Color table

```
          +---+---+---+----------------- Color code for part 1
          |   |   |   |   +---+---+---+-- Color code for part 0
          |   |   |   |   |   |   |   |
     MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+ Base address
  0     |FC3|FC2|FC1|FC0|BC3|BC2|BC1|BC0| Pattern Nos. 0 to 7
        +---+---+---+---+---+---+---+---+
  1     |FC3|FC2|FC1|FC0|BC3|BC2|BC1|BC0| Pattern Nos. 8 to 15
        +---+---+---+---+---+---+---+---+
  .     . . . . . . . . . . . . . . . . .     .     .    .   . .
  .     . . . . . . . . . . . . . . . . .     .     .    .   . .
        +---+---+---+---+---+---+---+---+
 31     |FC3|FC2|FC1|FC0|BC3|BC2|BC1|BC0| Pattern Nos. 248 to 255
        +---+---+---+---+---+---+---+---+
```

6. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 1.

```
     MSB  7   6   5   4   3   2   1   0   LSB
        +---+---+---+---+---+---+---+---+
R#5     |A14|A13|A12|A11|A10|A9 |A8 |A7 | Sprite attribute table
        +===+===+===+===+===+===+===+===+ base address register
R#11    | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15|
        +---+---+---+---+---+---+---+---+
        +---+---+---+---+---+---+---+---+
R#6     | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
        +---+---+---+---+---+---+---+---+  table base address
                                           register
```

Example of VRAM allocation in GRAPHIC 1 mode

```
00000H  +----------+
        | Sprite   |
        | Generator|
        | Table    |
        |  (1024   |
        |   bytes) |
00400H  +----------+
        | Pattern  |
        | Name     |     .+----------+ 00700H
        | Table    |    . | Sprite   |
        |  (768    |   .  | attribute|
        |   bytes) |  .   | table    |
00700H  +----------+.     |  (128    |
        |          |      |   bytes) |
        |          |      +----------+ 00780H
00800H  +----------+      | Color    |
        |          |.     | table    |
        |          | .    +(32 bytes)+ 007A0H
        |          |  .   |          |
        |          |   .  |          |
        |          |    .+----------+ 00800H
        |          |
        | Pattern  |
00C00H  | Generator|
        | Table    |
        |  (2048   |
        |   bytes) |
        |          |
        |          |
        |          |
01000H  +----------+
        .          .
        .          .         A maximum of 32 pages may be allocated in
        .          .         the same manner (using a 128K-byte VRAM).
        .          .
        |          |
        |          |
1FFFFH  +----------+
```

GRAPHIC 2 AND GRAPHIC 3 MODES

Characteristics

```
- Pattern size             : 8 dots (w) x 8 dots (h)
- Patterns                 : 768 types
- Screen pattern count     : 32 (w) x 24 (h) patterns
- Pattern colors           : 16 colors out of 512 colors (per screen)
- Sprite modes             : Sprite mode 1 (GRAPHIC 2)
                           : Sprite mode 2 (GRAPHIC 3)
- Vram area per screen     : 16K bytes

* The GRAPHIC 2 and GRAPHIC 3 modes are identical except for the
  sprite modes.
```

Controls

```
- Pattern font              : VRAM pattern generator table
- Screen pattern location   : VRAM pattern name table
- Pattern color codes 1 & 0: Can be specified as a group for each
                           : raster, in the VRAM color table
- Background color code    : Low-order four bits of R#7
- Sprites                  : VRAM sprite attribute table, VRAM sprite
                             pattern table.
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 0*| 0*| 0 | Mode register 0
           +---+---+---+---+---+-+-+-+-+---+
                                 |   +------ 0, 1 for GRAPHIC 2 mode
                                 +---------- 1, 0 for GRAPHIC 3 mode
           +---+---+---+---+---+---+---+---+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 2 mode or GRAPHIC 3 mode
** Indicates negative logic
```

All other bits are set accordingly.

2. Pattern Generator Table Settings

- The pattern generator table is an area that stores the pattern fonts.
- Each pattern has a number from PN0 to PN255; and since each group may have three members, 768 patterns may be specified.
- The font for each pattern is constructed from 8 bytes.
- Set the beginning (head) address of the pattern generator table in register R#4.

```
     MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
R#4     | 0 | 0 |A16|A15|A14|A13| 1 | 1 | Pattern generator table
        +---+---+---+---+---+---+---+---+ base address register
```

3. Color table settins

- The colors for pattern color 1 and pattern color 0 are set as a group of one raster.
- The color table corresponds to the pattern generator table on a one- to-one basis.
- Set the beginning (head) address of the color table in registers R#3 and R#10.

```
     MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
R#3     |A13| 1 | 1 | 1 | 1 | 1 | 1 | 1 | Color table address
        +===+===+===+===+===+===+===+===+
R#10    | 0 | 0 | 0 | 0 | 0 |A16|A15|A14| register
        +---+---+---+---+---+---+---+---+
```

4. Color register settings

```
     MSB  7   6   5   4   3   2   1   0   LSB
        +---+---+---+---+---+---+---+---+
R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop color
        +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ register
          |   |   |   |   |   |   |   |
          |   |   |   |   +---+---+---+-- Specifies backdrop color
          +---+---+---+----------------- Ignored
```

Pattern generator table

```
(X=1, O=0)
                            MSB 76543210 LSB      MSB 76543210 LSB
                      +-----   0 OOXXXOOO              OOOOOOOO
                      |        1 OXOOOXOO              OOOOOOOO
                      |        2 XOOOOOXO              OOOOOOOO
Pattern number 0   ---+        3 XOOOOOXO              OOOOOOOO
                      |        4 XXXXXXXO              OOOOOOOO
                      |        5 XOOOOOXO              OOOOOOOO
                      |        6 XOOOOOXO              OOOOOOOO
                      +-----   7 OOOOOOOO              OOOOOOOO
                      +-----   8 XXXXXXOO              OOOOOOOO
                      |        9 XOOOOOXO              OOOOOOOO
                      |       10 XOOOOOXO              OOOOOOOO
Pattern number 0   ---+       11 XXXXXXOO              OOOOOOOO
                      |       12 XOOOOOXO              OOOOOOOO
                      |       13 XOOOOOXO              OOOOOOOO
                      |       14 XXXXXXOO              OOOOOOOO
                      +-----  15 OOOOOOOO              OOOOOOOO
                         .     .      .                    .
                         .     .      .                    .
                         .     .      .                    .
                      +-----2040 XOXOXOXO              OOOOOOOO
                      |     2041 OXOXOXOX              OOOOOOOO
                      |     2042 XOXOXOXO              OOOOOOOO
Pattern number 255 ---+     2043 OXOXOXOX              OOOOOOOO
                      |     2044 XOXOXOXO              OOOOOOOO
                      |     2045 OXOXOXOX              OOOOOOOO
                      |     2046 XOXOXOXO              OOOOOOOO
                      +-----2047 OXOXOXOX              OOOOOOOO
                                                       |   ||  |
                            Color pattern 1 -----------+---+|  |
                            Color pattern 0 ---------------+---+


     Pattern generator       Color table
           table

   0 |             |       |             |
     +-------------+       +-------------+ -+
     | 256 patterns|       | 256 colors  |  |
     | for upper   |       | for upper   |  |
     | third of    |       | third of    |  |.
     | screen      |       | screen      |  | ..  +-----------------+
     | (2048 bytes)|       | (2048 bytes)|  |    ..|     Upper       |
2048 +-------------+       +-------------+ -+     |     Third       |
     | 256 patterns|       | 256 colors  |  |     +-----------------+
     | for middle  |       | for middle  |  |.....|     Middle      |
     | third of    |       | third of    |  |     |     Third       |
     | screen      |       | screen      |  |     +-----------------+
     | (2048 bytes)|       | (2048 bytes)|  |     |     Lower       |
4096 +-------------+       +-------------+ -+    .|     Third       |
     | 256 patterns|       | 256 colors  |  |   . +-----------------+
     | for lower   |       | for lower   |  | ..     Display screen
     | third of    |       | third of    |  |.
     | screen      |       | screen      |  |
```

```
     | (2048 bytes)|        | (2048 bytes)|  |
6144 +-------------+        +-------------+ -+
     |             |        |             |
```

5. Pattern name table settings

- The pattern name table is composed of one byte for each screen pattern. Each byte specifies a unique pattern.
- The upper, middle, and lower parts of the screen can be used as three different parts, for a total of 768 patterns.

Pattern name table

```
+-------------------------------------------------------+
| ( 0, 0)    Pattern display area for upper    (31, 0) |
|              third of screen (256 bytes)             |
| ( 0, 7)                                      (31, 7) |
+-------------------------------------------------------+
| ( 0, 8)    Pattern display area for middle   (31, 8) |
|              third of screen (256 bytes)             |
| ( 0,15)                                      (31,15) |
+-------------------------------------------------------+
| ( 0,16)    Pattern display area for lower    (31,16) |
|              third of screen (256 bytes)             |
| ( 0,23)                                      (31,23) |
+-------------------------------------------------------+
```

- Set the beginning (head) address of the pattern name table in register R#2.

```
        MSB  7   6   5   4   3   2   1   0   LSB
           +---+---+---+---+---+---+---+---+
   R#2     | 0 |A16|A15|A14|A13|A12|A11|A10| Pattern name table
           +---+---+---+---+---+---+---+---+ base address register
```

Pattern name table

```
|         |
+---------+ Base address  --------------+
| ( 0, 0) |   0                         |
+---------+                             |
| ( 1, 0) |   1                         +-- For upper third
+---------+                             |    of screen
.         .   .                         |
+---------+                             |
| (31, 7) |  255                        |
+---------+       --------------------+
| ( 0, 8) |  256                        |
+---------+                             +-- For middle third
.         .   .                         |    of screen
+---------+                             |
| (31,15) |  511                        |
+---------+       --------------------+
| ( 0,16) |  512                        |
+---------+                             +-- For lower third
.         .   .                         |    of screen
+---------+                             |
| (31,23) | 767 (byte)                  |
+---------+       --------------------+
```

6. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites pertaining to GRAPHIC 2 MODE, see the section on SPRITE MODE 1, and for details about sprites pertaining to GRAPHIC 3 mode, see the section on SPRITE MODE 2.

```
      MSB  7   6   5   4   3   2   1   0   LSB
          +---+---+---+---+---+---+---+---+
R#5       |A14|A13|A12|A11|A10|A9 |A8 |A7 | Sprite attribute table
          +===+===+===+===+===+===+===+===+ base address register
R#11      | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15|
          +---+---+---+---+---+---+---+---+
          +---+---+---+---+---+---+---+---+
R#6       | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
          +---+---+---+---+---+---+---+---+  table base address
                                             register
```

Example of VRAM allocation in GRAPHIC 2 mode

```
00000H  +-----------+
        | Pattern   |
        | Generator|
        | Table     |
        | Upper     |
00800H  +-----------+
        | Pattern   |
        | Generator|
        | Table     |
        | Middle    |
01000H  +-----------+
        | Pattern   |     .+-----------+ 01C00H
        | Generator|    . | Sprite    |
        | Table     |   .  | generator|
        | Lower     |  .   | table     |
01800H  +-----------+.     |           |
        |           |      |           |
        |           |      +-----------+ 01C00H
02000H  +-----------+      | Sprite    |
        | Color     |.     | attribute|
        | Table     |  .   | table     |
        | Upper     |   .  +-----------+ 01C80H
02800H  +-----------+   .  |           |
        | Color     |    . |           |
        | Table     |     .|           |
        | Middle    |      +-----------+ 02000H
        +-----------+
03000H  | Color     |
        | Table     |
        | Lower     |
03800H  +-----------+
        | Pattern   |
        | Name      |
        | Table     |
03B00H  +-----------+
        | Table     |
04000H  +-----------+
        |           |
        |           |
        |           |
        .           .
        .           .        A maximum of 8 pages may be allocated in
        .           .        the same manner (using a 128K-byte VRAM).
        .           .
        |           |
        |           |
1FFFFH  +-----------+
```

Example of VRAM allocation in GRAPHIC 3 mode

```
00000H  +-----------+
        | Pattern   |
        | Generator|
        | Table     |
        | Upper     |
00800H  +-----------+
        | Pattern   |
        | Generator|
        | Table     |
        | Middle    |
01000H  +-----------+
        | Pattern   |     .+-----------+ 01C00H
        | Generator|    . | Sprite    |
        | Table     |   .  | generator|
        | Lower     |  .   | table     |
01800H  +-----------+.     |           |
        |           |      |           |
        |           |      +-----------+ 01C00H
02000H  +-----------+      | Sprite    |
        | Color     |.     | color     |
        | Table     |.     | table     |
        | Upper     | .    +-----------+ 01E00H
02800H  +-----------+ .    | Sprite    |
        | Color     |  .   | attribute|
        | Table     |  .   | table     |
        | Middle    |   .  +-----------+ 01E80H
        +-----------+   .  |           |
03000H  | Color     |    .|           |
        | Table     |    .+-----------+ 02000H
        | Lower     |
03800H  +-----------+
        | Pattern   |
        | Name      |
        | Table     |
03B00H  +-----------+
        | Table     |
04000H  +-----------+
        |           |
        |           |
        |           |
        .           .
        .           .        A maximum of 8 pages may be allocated in
        .           .        the same manner (using a 128K-byte VRAM).
        .           .
        |           |
        |           |
1FFFFH  +-----------+
```

GRAPHIC 4 MODE

Characteristics

```
- Bit-mapped Graphics Mode
- Screen size                : 256 (w) x 212 (h) dots
                             : 256 (w) x 192 (h) dots
- Screen colors              : 16 colors out of 512 colors (per screen)
- Sprite mode                : Sprite mode 2
- VRAM area per screen       : 32K bytes
```

Controls

```
- Graphics                   : VRAM pattern name table
- Background color code      : Low-order four bits of R#7
- Sprites                    : VRAM sprite attribute table, VRAM sprite
                               pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 0*| 1*| 1*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 4 mode
** Indicates negative logic
```

In GRAPHIC 4 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every two dots on the screen. A color can be assigned for each dot from a collection of 16 colors out of 512 colors.

```
+----> X
|  +-------+-------+.................+---------+---------+
|  |(0,  0)|(1,  0)|                 |(254,  0)|(255,  0)|
v  +-------+-------+                 +---------+---------+
Y  |(0,  1)|                                   |(255,  1)|
   +-------+            +----------+           +---------+
   |                    |   (X,Y)  |                     |
   .                    +----------+                     .
   .                                                     .
   .                                                     .
   |                                                     |
   |                                                     |
   |                                                     |
   |                                                     |
   +-------+                                   +---------+
   |(0,191)|                                   |(255,191)|
   +-------+-----------------------------------+---------+   LN = 0

   |                                                     |
   +-------+                                   +---------+
   |(0,211)|                                   |(255,211)|
   +-------+-----------------------------------+---------+   LN = 1
```

- Set the beginning (head) address of the pattern name table in register R#2.

```
      MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#2     | 0 |A16|A15| 1 | 1 | 1 | 1 | 1 | Pattern name table
         +---+-+-+-+-+---+---+---+---+---+ base address register
               |   |
               +---+---------------------- These two bits specify
                                           the page to display
```

Pattern name table

```
 MSB     7   6   5   4   3   2   1   0    LSB
       +----+---+---+---+---+---+---+---+
    1  |    (  0,  0)   |   (  1,  0)   |  Base address
       +----+---+---+---+---+---+---+---+
    2  |    (  2,  0)   |   (  3,  0)   |  Set the color code for each
       +----+---+---+---+---+---+---+---+  dot in this table
       .                                .
       .                                .
       .                                .
       +----+---+---+---+---+---+---+---+
  127  |    (254,  0)   |   (255,  0)   |
       +----+---+---+---+---+---+---+---+
  128  |    (  0,  1)   |   (  1,  1)   |
       +----+---+---+---+---+---+---+---+
       .                                .
       .                                .
       .                                .
       +----+---+---+---+---+---+---+---+
27134  |    (252,211)   |   (253,211)   |
       +----+---+---+---+---+---+---+---+
27135  |    (254,211)   |   (255,211)   |
       +----+---+---+---+---+---+---+---+
```

3. Color register settings

```
        MSB  7   6   5   4   3   2   1   0   LSB
           +---+---+---+---+---+---+---+---+
   R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
           +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ color register
             |   |   |   |   |   |   |   |
             |   |   |   |   +---+---+---+-- Specifies backdrop color
             |   |   |   |                   code
             +---+---+---+------------------ Ignored
```

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

```
          MSB  7   6   5   4   3   2   1   0  LSB
              +---+---+---+---+---+---+---+---+
R#5           |A14|A13|A12|A11|A10| 1 | 1 | 1 | Sprite attribute table
              +===+===+===+===+===+===+===+===+
R#11          | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15| base address register
              +---+---+---+---+---+---+---+---+

              +---+---+---+---+---+---+---+---+
R#6           | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
              +---+---+---+---+---+---+---+---+ table base address
                                                register
```

Example of VRAM allocation in GRAPHIC 4 mode

```
00000H  +-----------+
        |           |
        |           |
        |           |
02000H  | Pattern   |
        |  name     |
        |  table    |
        |           |
04000H  |           |
        |           |
        |           |
        | 192 lines|
06000H  +-----------+
        |           |     .+-----------+ 07000H
        |           |    . | Sprite    |
        | 212 lines| .-   | generator|
06A00H  +-----------+.     | table     |
        |           |      |  (2048    |
07000H  +-----------+      |   bytes)  |
        |           |      +-----------+ 07800H
        |           |      | Sprite    |
        |           |      | color     |
        |           |      | table     |
        |           |      |  (512     |
        |           |      |   bytes)  |
        |           |      +-----------+ 07A00H
        |           |      | Sprite    |
        |           |      | attribute|
        |           |      | table     |
08000H  +-----------+.     |  (128     |
        .           .  .   |   bytes   |
        .           .   .  +-----------+ 07A80H
        .           .    . |           |
        .           .     .+-----------+ 08000H
        .           .
        |           |
        |           |        A maximum of 4 pages may be allocated in
1FFFFH  +-----------+        the same manner (using a 128K-byte VRAM).
```

GRAPHIC 5 MODE

Characteristics

```
- Bit-mapped Graphics Mode
- Screen size                : 512 (w) x 212 (h) dots
                             : 512 (w) x 192 (h) dots
- Screen colors              : 4 colors out of 512 colors (per screen)
- Sprite mode                : Sprite mode 2
- VRAM area per screen       : 32K bytes
```

Controls

```
- Graphics                   : VRAM pattern name table
- Background color code      : Low-order four bits of R#7
- Sprites                    : VRAM sprite attribute table, VRAM sprite
                               pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 1*| 0*| 0*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 4 mode
** Indicates negative logic
```

In GRAPHIC 5 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every four dots on the screen. A color can be assigned for each dot from a selection of 4 colors out of 512 colors.

```
+----> X
|   +-------+-------+.................+---------+---------+
|   |(0,  0)|(1,  0)|                 |(510,  0)|(511,  0)|
v   +-------+-------+                 +---------+---------+
Y   |(0,  1)|                                   |(511,  1)|
    +-------+          +---------+              +---------+
    |                  |  (X,Y)  |                        |
    .                  +---------+                        .
    .                                                     .
    .                                                     .
    |                                                     |
    |                                                     |
    |                                                     |
    |                                                     |
    +-------+                                   +---------+
    |(0,191)|                                   |(511,191)|
    +-------+-----------------------------------+---------+   LN = 0

    |                                                     |
    +-------+                                   +---------+
    |(0,211)|                                   |(511,211)|
    +-------+-----------------------------------+---------+   LN = 1
```

- Set the beginning (head) address of the pattern name table in register R#2.

```
      MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#2     | 0 |A16|A15| 1 | 1 | 1 | 1 | 1 | Pattern name table
         +---+-+-+-+-+---+---+---+---+---+ base address register
               |   |
               +---+--------------------- These two bits specify
                                          the page to display
```

Pattern name table

```
  MSB     7   6   5   4   3   2   1   0    LSB
        +----+---+---+---+---+---+---+---+
    1   | ( 0, 0)|( 1, 0)|( 2, 0)|( 3, 0)|  Base address
        +----+---+---+---+---+---+---+---+
    2   | ( 4, 0)|( 5, 0)|( 6, 0)|( 7, 0)|  Set the color code for each
        +----+---+---+---+---+---+---+---+  dot in this table
        .                                .
        .                                .
        .                                .
        +----+---+---+---+---+---+---+---+
  127   | (508,0)|(509,0)|(510,0)|(511,0)|
        +----+---+---+---+---+---+---+---+
  128   | ( 0, 1)|( 1, 1)|( 2, 1)|( 3, 1)|
        +----+---+---+---+---+---+---+---+
        .                                .
        .                                .
        .                                .
        +----+---+---+---+---+---+---+---+
27135   |    |   |   |   |   |  (511,211)|
        +----+---+---+---+---+---+---+---+
```

3. Color register settings

```
         MSB  7   6   5   4   3   2   1   0   LSB
            +---+---+---+---+---+---+---+---+
    R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
            +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ color register
              |   |   |   |   |   |   |   |
              |   |   |   |   +---+---+---+-- Specifies backdrop color
              |   |   |   |                   code
              +---+---+---+------------------ Ignored
```

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

```
         MSB  7   6   5   4   3   2   1   0  LSB
            +---+---+---+---+---+---+---+---+
    R#5     |A14|A13|A12|A11|A10| 1 | 1 | 1 | Sprite attribute table
            +===+===+===+===+===+===+===+===+
    R#11    | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15| base address register
            +---+---+---+---+---+---+---+---+

            +---+---+---+---+---+---+---+---+
    R#6     | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
            +---+---+---+---+---+---+---+---+ table base address
                                              register
```

5. Hardware tiling function

- In GRAPHIC 5 mode, a hardware tiling function processes the sprite and background colors. For these colors, you can specify four bits; however, of these four bits, the higher-order two bits specify the color code of the even dots, and the lower-order two bits specify the color code of the odd dots of the x-coordinage (0 to 511).
- In GRAPHIC 5 mode, the size of one dot of a sprite is approximately twice that of a graphics dot; however, when this tiling function is used, one dot of a sprite may be displayed in two colors.
- The even and odd dots of the background colors may also be specified in the same manner.

```
       +-------- Even dots (0, 2, . . . 510)
       |    +--- Odd dots  (1, 3, . . . 511)
       |    |
      ++--+-+-+
      |   |   |  Two graphics dots
      +---+---+

      +-------+
      |       |  A sprite dot
      +-------+

      +---+---+
      |   |XXX|
      ++--++--+
       |   |
      +-+ +-+
       | | | |
      +-+-+-+-+
 MSB  |/|/|/|/|  LSB   Bit specifying sprite colors (4 bits)
      +-+-+-+-+
```

Example of VRAM allocation in GRAPHIC 5 mode

```
00000H  +-----------+
        |           |
        | Pattern   |
        |  name     |
02000H  | table     |
        |           |
        |           |
        |           |
04000H  |           |
        |           |
        |           |
        | 192 lines|
06000H  +-----------+
        |           |     .+-----------+ 07000H
        |           |    . | Sprite    |
        | 212 lines| .-   | generator|
06A00H  +-----------+.     | table     |
        |           |      |  (2048    |
07000H  +-----------+      |   bytes) |
        |           |      +-----------+ 07800H
        |           |      | Sprite    |
        |           |      | color     |
        |           |      | table     |
        |           |      |  (512     |
        |           |      |   bytes) |
        |           |      +-----------+ 07A00H
        |           |      | Sprite    |
        |           |      | attribute|
        |           |      | table     |
        |           |      |  (128     |
        |           |      |   bytes   |
        |           |      +-----------+ 07A80H
        |           |      |           |
08000H  +-----------+.....+-----------+ 08000H
        .           .
        .           .
        .           .        A maximum of 4 pages may be allocated in
        .           .        the same manner (using a 128K-byte VRAM).
        .           .
        |           |
        |           |
1FFFFH  +-----------+
```

GRAPHIC 6 MODE

Characteristics

```
- Bit-mapped Graphics Mode
- Screen size              : 512 (w) x 212 (h) dots
                           : 512 (w) x 192 (h) dots
- Screen colors            : 16 colors out of 512 colors (per screen)
- Sprite mode              : Sprite mode 2
- VRAM area per screen     : 128K bytes (Two screens)

* To use this mode, the VRAM must have 128K bytes.
```

Controls

```
- Graphics                 : VRAM pattern name table
- Background color code    : Low-order four bits of R#7
- Sprites                  : VRAM sprite attribute table, VRAM sprite
                             pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 1*| 0*| 1*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 4 mode
** Indicates negative logic
```

In GRAPHIC 6 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every two dots on the screen. A color can be assigned for each dot from a selection of 16 colors out of 512 colors.

```
+----> X
|   +-------+-------+..................+---------+---------+
|   |(0,  0)|(1,  0)|                  |(510,  0)|(511,  0)|
v   +-------+-------+                  +---------+---------+
Y   |(0,  1)|                                    |(511,  1)|
    +-------+           +----------+             +---------+
    |                   |   (X,Y)  |                       |
    .                   +----------+                       .
    .                                                      .
    .                                                      .
    |                                                      |
    |                                                      |
    |                                                      |
    |                                                      |
    +-------+                                    +---------+
    |(0,191)|                                    |(511,191)|
    +-------+------------------------------------+---------+   LN = 0

    |                                                      |
    +-------+                                    +---------+
    |(0,211)|                                    |(511,211)|
    +-------+------------------------------------+---------+   LN = 1
```

- Set the beginning (head) address of the pattern name table in register R#2.

```
      MSB  7   6   5   4   3   2   1   0  LSB
         +---+---+---+---+---+---+---+---+
 R#2     | 0 | 0 |A16| 1 | 1 | 1 | 1 | 1 | Pattern name table
         +---+-+-+-+-+---+---+---+---+---+ base address register
                   |
                   +------- Specifies the page to display (In G6 and
                            G7 modes, the position of A16 differs).
```

Pattern name table

```
  MSB     7   6   5   4   3   2   1   0    LSB
       +----+---+---+---+---+---+---+---+
    1  |    (  0,  0)   |   (  1,  0)   |  Base address
       +----+---+---+---+---+---+---+---+
    2  |    (  2,  0)   |   (  3,  0)   |  Set the color code for each
       +----+---+---+---+---+---+---+---+  dot in this table
       .                                .
       .                                .
       .                                .
       +----+---+---+---+---+---+---+---+
  255  |    (510,  0)   |   (511,  0)   |
       +----+---+---+---+---+---+---+---+
  256  |    (  0,  1)   |   (  1,  1)   |
       +----+---+---+---+---+---+---+---+
       .                                .
       .                                .
       .                                .
       +----+---+---+---+---+---+---+---+
54270  |    (508,211)   |   (509,211)   |
       +----+---+---+---+---+---+---+---+
54271  |    (510,211)   |   (511,211)   |
       +----+---+---+---+---+---+---+---+
```

3. Color register settings

```
        MSB  7   6   5   4   3   2   1   0   LSB
           +---+---+---+---+---+---+---+---+
   R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
           +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ color register
             |   |   |   |   |   |   |   |
             |   |   |   |   +---+---+---+-- Specifies backdrop color
             |   |   |   |                   code
             +---+---+---+------------------ Ignored
```

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

```
     MSB  7   6   5   4   3   2   1   0  LSB
        +---+---+---+---+---+---+---+---+
R#5     |A14|A13|A12|A11|A10| 1 | 1 | 1 | Sprite attribute table
        +===+===+===+===+===+===+===+===+
R#11    | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15| base address register
        +---+---+---+---+---+---+---+---+

        +---+---+---+---+---+---+---+---+
R#6     | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
        +---+---+---+---+---+---+---+---+ table base address
                                          register
```

Example of VRAM allocation in GRAPHIC 6 mode

```
00000H  +----------+
        |          |
        | Pattern  |
        | Name     |
        | table    |
        |          |
        |          |
        |          |
04000H  |          |
        |          |
        |          |
        |          |
        |          |
        |          |
08000H  |          |
        |          |
        |          |     +----------+ 0F000H
        |          |    .| Sprite   |
        | 192 line |    .| generator|
0C000H  +----------+    .| table    |
        | 212 line |   . |  (2048   |
0D400H  +----------+   . |   bytes) |
        |          |   . +----------+ 0F800H
        |          |   . | Sprite   |
        |          |  .  | color    |
        |          |  .  | table    |
        |          |  .  |  (512    |
        |          | .   |   bytes) |
        |          | .   +----------+ 0FA00H
        |          | .   | Sprite   |
        |          | .   | attribute|
        |          |.    | table    |
        |          |.    |  (128    |
        |          |.    |   bytes  |
0F000H  +----------+     +----------+ 0FA80H
        |          |     |          |
08000H  +----------+.....+----------+ 10000H
        .          .
        .          .
        .          .        A maximum of 2 pages may be allocated in
        .          .        the same manner (using a 128K-byte VRAM).
        .          .
        |          |
        |          |
1FFFFH  +----------+
```

GRAPHIC 7 MODE

Characteristics

```
- Bit-mapped Graphics Mode
- Screen size              : 256 (w) x 212 (h) dots
                           : 256 (w) x 192 (h) dots
- Screen colors            : 256 colors (per screen)
- Sprite mode              : Sprite mode 2
- VRAM area per screen     : 128K bytes (Two screens)

* To use this mode, the VRAM must have 128K bytes.
```

Controls

```
- Graphics                 : VRAM pattern name table
- Background color code    : Low-order four bits of R#7
- Sprites                  : VRAM sprite attribute table, VRAM sprite
                             pattern table
```

Initial Settings

1. Mode and Register Settings

```
        MSB  7   6   5   4   3   2   1   0  LSB
           +---+---+---+---+---+---+---+---+
   R#0     | 0 |DG |IE2|IE1| 1*| 1*| 1*| 0 | Mode register 0
           +===+===+===+===+===+===+===+===+
   R#1     | 0 |BL |IE0| 0*| 0*| 0 |SI |MAG| Mode register 1
           +===+===+===+===+===+===+===+===+
   R#8     |MS |LP |TP |CB |VR | 0 |SPD|BW | Mode register 2
           +===+===+===+===+===+===+===+===+
   R#9     |LN | 0 |S1 |S0 |IL |EO **NT|DC | Mode register 3
           +---+---+---+---+---+---+---+---+
*  Examples of settings in GRAPHIC 4 mode
** Indicates negative logic
```

In GRAPHIC 7 mode, if LN is set to 1, the screen height is 212 dots, and if LN is set to 0, the screen height is 192 dots.
All other bits are set accordingly.

2. Pattern name table settings

- The pattern name table is composed of one byte for every dot on the screen. A color can be assigned for each dot from a selection of 256 colors.

```
+----> X
|   +-------+-------+.................+---------+---------+
|   |(0,   0)|(1,   0)|                 |(254,   0)|(255,   0)|
v   +-------+-------+                 +---------+---------+
Y   |(0,   1)|                                  |(255,   1)|
    +-------+          +----------+          +---------+
    |                  |   (X,Y)   |                    |
    .                  +----------+                    .
    .                                                  .
    .                                                  .
    |                                                  |
    |                                                  |
    |                                                  |
    |                                                  |
    +-------+                                +---------+
    |(0,191)|                                |(255,191)|
    +-------+--------------------------------+---------+   LN = 0

    |                                                  |
    +-------+                                +---------+
    |(0,211)|                                |(255,211)|
    +-------+--------------------------------+---------+   LN = 1
```

- Set the beginning (head) address of the pattern name table in register R#2.

```
      MSB   7    6    5    4    3    2    1    0   LSB
          +---+---+---+---+---+---+---+---+
 R#2      | 0 | 0 |A16| 1 | 1 | 1 | 1 | 1 | Pattern name table
          +---+-+-+-+-+---+---+---+---+---+ base address register
                    |
                    +---- Specifies the page to display; in the G6
                          and G7 modes only, the location of the A16
                          bit differs.
```

Pattern name table

```
 MSB     7   6   5   4   3   2   1   0    LSB
       +----+---+---+---+---+---+---+---+
    1  |    |   | (   0,   0)   |   |   |  Base address
       +----+---+---+---+---+---+---+---+
    2  |    |   | (   1,   0)   |   |   |  Set the color code for each
       +----+---+---+---+---+---+---+---+  dot in this table
       .                                .
       +----+---+---+---+---+---+---+---+
       |   GREEN    |    RED    | BLUE  |
       +----+---+---+---+---+---+---+---+
       .                                .
       +----+---+---+---+---+---+---+---+
  255  |    |   | ( 255,   0)   |   |   |
       +----+---+---+---+---+---+---+---+
  256  |    |   | (   0,   1)   |   |   |
       +----+---+---+---+---+---+---+---+
       .                                .
       .                                .
       .                                .
       +----+---+---+---+---+---+---+---+
54270  |    |   | ( 254, 211)   |   |   |
       +----+---+---+---+---+---+---+---+
54271  |    |   | ( 255, 211)   |   |   |
       +----+---+---+---+---+---+---+---+
```

3. Color register settings

```
        MSB  7   6   5   4   3   2   1   0   LSB
           +---+---+---+---+---+---+---+---+
   R#7     |TC3|TC2|TC1|TC0|BD3|BD2|BD1|BD0| Text color/Back drop
           +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ color register
             |   |   |   |   |   |   |   |
             +---+---+---+---+---+---+---+-- Specifies backdrop color
                                             code
```

4. Sprite settings

- Set the beginning (head) address of the sprite attribute table in registers R#5 and R#11; and set the beginning (head) address of the sprite pattern generator table in register R#6. For details about sprites, see the section on SPRITE MODE 2.

```
          MSB  7   6   5   4   3   2   1   0  LSB
              +---+---+---+---+---+---+---+---+
R#5           |A14|A13|A12|A11|A10| 1 | 1 | 1 | Sprite attribute table
              +===+===+===+===+===+===+===+===+
R#11          | 0 | 0 | 0 | 0 | 0 | 0 |A16|A15| base address register
              +---+---+---+---+---+---+---+---+

              +---+---+---+---+---+---+---+---+
R#6           | 0 | 0 |A16|A15|A14|A13|A12|A11| Sprite pattern generator
              +---+---+---+---+---+---+---+---+ table base address
                                                register
```

Example of VRAM allocation in GRAPHIC 7 mode

```
00000H  +----------+
        |          |
        | Pattern  |
        | Name     |
        | table    |
        |          |
        |          |
        |          |
04000H  |          |
        |          |
        |          |
        |          |
        |          |
        |          |
08000H  |          |
        |          |
        |          |     +----------+ 0F000H
        |          |    .| Sprite   |
        | 192 line |    .| generator|
0C000H  +----------+    .| table    |
        | 212 line |   . |  (2048   |
0D400H  +----------+   . |   bytes) |
        |          |   . +----------+ 0F800H
        |          |   . | Sprite   |
        |          |  .  | color    |
        |          |  .  | table    |
        |          |  .  |  (512    |
        |          | .   |   bytes) |
        |          | .   +----------+ 0FA00H
        |          | .   | Sprite   |
        |          | .   | attribute|
        |          |.    | table    |
        |          |.    |  (128    |
        |          |.    |   bytes  |
0F000H  +----------+     +----------+ 0FA80H
        |          |     |          |
08000H  +----------+.....+----------+ 10000H
        .          .
        .          .
        .          .        A maximum of 2 pages may be allocated in
        .          .        the same manner (using a 128K-byte VRAM).
        .          .
        |          |
        |          |
1FFFFH  +----------+
```

COMMANDS

1. Types of Commands

It is very easy to use MSX-VIDEO commands to perform functions such as LINE and PSET for use with graphics, and for transferring parts of the screen.

Command Name	Destination	Source	Late	Mnemonic	CM3	CM2	CM3	CM0
High-speed move	VRAM	CPU	Byte	HMMC	1	1	1	1
	VRAM	VRAM	Byte	YMMM	1	1	1	0
	VRAM	VRAM	Byte	HMMM	1	1	0	1
	VRAM	VDP	Byte	HMMV	1	1	0	0
Logical move	VRAM	CPU	Dot	LMMC	1	0	1	1
	CPU	VRAM	Dot	LMCM	1	0	1	0
	VRAM	VRAM	Dot	LMMM	1	0	0	1
	VRAM	VDP	Dot	LMMV	1	0	0	0
Line	VRAM	VDP	Dot	LINE	0	1	1	1
Search	VRAM	VDP	Dot	SRCH	0	1	1	0
Pset	VRAM	VDP	Dot	PSET	0	1	0	1
Point	VDP	VRAM	Dot	POINT	0	1	0	0
Invalid					0	0	1	1
Invalid					0	0	1	0
Invalid					0	0	0	1
Stop					0	0	0	0

- Commands are executed in the MSX-VIDEO by writing the data into R#46 (the Command Register, hereafter abbreviated CMR), and setting bit 0 of status register S#2 (CE/Command Execute) to 1. Before this can be done, the necessary parameters must first have been set in registers R#32 to R#45.
- When the command execution is complete, CE is set to 0.
- To abort a command while it is being executed, execute a STOP.
- The results of command execution are only guaranteed during bit map mode (Graph4 to Graph7).

2. Page Concept

The parameters used for the MSX-VIDEO are all x-y coordinates. In other words, the internal command processor of the MSX-VIDEO accesses the entire VRAM area as x-y coordinates of the display mode.

When a screen is to be displayed, 212 lines of the same page are displayed (selected by R#23). To select the page to be displayed, use R#2.

When a command is being executed, the contents of the display screen are ignored.

The display modes and their relationships to the coordinates are shown in the table below.

```
         GRAPH 4                Address            GRAPH 5
+--------------------------+  00000H  +--------------------------+
|(0,0)              (255,0)|    |     |(0,0)              (511,0)|
|          Page 0          |    |     |          Page 0          |
|(0,255)          (255,255)|    |     |(0,255)          (511,255)|
+--------------------------+  08000H  +--------------------------+
|(0,256)          (255,256)|    |     |(0,256)          (511,256)|
|          Page 1          |    |     |          Page 1          |
|(0,511)          (255,511)|    |     |(0,511)          (511,511)|
+--------------------------+  10000H  +--------------------------+
|(0,512)          (255,512)|    |     |(0,512)          (511,512)|
|          Page 2          |    |     |          Page 2          |
|(0,767)          (255,767)|    |     |(0,767)          (511,767)|
+--------------------------+  18000H  +--------------------------+
|(0,768)          (255,768)|    |     |(0,768)          (511,768)|
|          Page 3          |    |     |          Page 3          |
|(0,1023)        (255,1023)|    |     |(0,1023)        (511,1023)|
+--------------------------+  1FFFFH  +--------------------------+
```

```
         GRAPH 7                Address            GRAPH 6
+--------------------------+  00000H  +--------------------------+
|(0,0)              (255,0)|    |     |(0,0)              (511,0)|
|          Page 0          |    |     |          Page 0          |
|(0,255)          (255,255)|    |     |(0,255)          (511,255)|
+--------------------------+  10000H  +--------------------------+
|(0,256)          (255,256)|    |     |(0,256)          (511,256)|
|          Page 1          |    |     |          Page 1          |
|(0,511)          (255,511)|    |     |(0,511)          (511,511)|
+--------------------------+  1FFFFH  +--------------------------+
```

3. Logical Operations

When the LINE, PSET and LOGICAL MOVE commands are executed on the MSX-VIDEO, the operations may be performed on the color on the screen. To do logical operations on the MSX-VIDEO, write the lower four bits of R#46 (Command register) simultaneously when you specify the command.

Summary of Logical Operations

Name	Operation	LO3	LO2	LO1	LO0
IMP	DC=SC	0	0	0	0
AND	SC*DC	0	0	0	1
OR	SC+DC	0	0	1	0
EOR	$\overline{SC}$*DC+SC*$\overline{DC}$	0	0	1	1
NOT	DC=$\overline{SC}$	0	1	0	0
---		0	1	0	1
---		0	1	1	0
---		0	1	1	1
TIMP	if SC=0 then DC=DC else DC=SC	1	0	0	0
TAND	if SC=0 then DC=DC else SC*DC	1	0	0	1
TOR	if SC=0 then DC=DC else SC+DC	1	0	1	0
TEOR	if SC=0 then DC=DC else $\overline{SC}$*DC+SC*$\overline{DC}$	1	0	1	1
TNOT	if SC=0 then DC=DC else DC=$\overline{SC}$	1	1	0	0
---		1	1	0	1
---		1	1	1	0
---		1	1	1	1

* SC = Source Color code
* DC = Destination Color code
* EOR = Exclusive OR

4. Explanation of Commands

4.1 HMMC (High-speed move CPU to VRAM)

The HMMC command transfers data from the CPU to the Video or expansion RAM in a specified rectangular area (in x-y coordinates) via the MSX-VIDEO.

Since the data to be transferred is done in units of one byte, there is a limitation, according to the display mode, on the value for x.

```
        Video or Expansion RAM
+---------------------------------------+
|                                       |
|   (DX,DY)                             |     +------+       +------+
|      +------------+   -> DIX          |     |      |       |      |
|      |     NX     |                   |     |MSX-  |       | CPU  |
|      |NY          | <-----------------+-----+VIDEO|  <---  |      |
|      +------------+                   |     |      |       |      |
|      |                                |     +------+       +------+
|      v                                |
+---------------------------------------+
```

4.1.1 HMMC Execution Order

1. First, set the necessary parameters in the command register of the MSX-VIDEO.

MXD:	Select destination memory 0: Video RAM 1: Expansion RAM
DX:	Basic x-coordinate of destination (0 to 511) *1
DY:	Basic y-coordinate of destination (0 to 1023)
NX:	Dots to move in x-direction (0 to 511) *1
NY:	Dots to move in x-direction (0 to 1023)

*1 Note that in the G4 and G6 modes, the lower one bit, and in the G5 mode, the lower two bits are lost.

DIX:	Direction for NX from x-coordinate of destination 0: Right 1: Left
DIY:	Direction for NY from y-coordinate of destination 0: Down 1: Up
CLR:	(R#44: Color register) First byte of data to be transferred

2. After you specify the above data, execute the command by writing 1 1 1 1 0 0 0 0 B into the CMR (R#46: Command register).

3. While checking TR and CE in Status Register #2, send the second byte and all bytes following into the CLR register.

4.1.2 Setting up the HMMC register